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DRIVE SYSTEM FOR PATIENT LIFT

Abstract

A drive system for a patient lift. The drive system comprises a drum configured to control the vertical movement of a patient support mounting device of the patient lift via a load bearing member, at least one motor adapted to drive the drum, each motor being connected to an motor shaft gear, and a transmission connecting the motor and the drum, the transmission being adapted to transfer torque from the motor to the drum.

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Background/Summary

RELATED APPLICATION DATA [0001] This application is a continuation application of U.S. patent application Ser. No. 18/021,471, filed Feb. 15, 2023, which is a national stage application of PCT/EP2021/072703, filed Aug. 16, 2021, which claims priority under 35 U.S.C. § 119 to Swedish Application No. 2050957-6, filed Aug. 17, 2020, the entire content of each of these applications is incorporated herein by reference.

TECHNOLOGY FIELD
[0002] The present invention relates to a drive system for a patient lift. The present invention further relates to a patient lift comprising such a drive system. Further the present invention relates to a method for controlling a torque exerted by each of at least two motors comprised in a drive system.
BACKGROUND
[0003] Patient lifts, also referred to as patient hoists, are commonly used to raise, lower and transfer patients who are disabled or who otherwise have mobility problems. Two common types of patient lifts are stanchion-mounted lifts, also known as floor lifts and ceiling lifts. Floor lifts often have a hoist assembly which may be disposed at the upper end of a stanchion. The stanchion has a wheeled base, which allows for the lift to be moved along the ground to different locations
[0004] A lifting member which may be in the form of a spreader bar, such as a two-point attachment spreader bar, a three-point attachment spreader bar, a four-point attachment spreader bar, a five-point attachment spreader bar or a powered spreader bar for adjusting the angle of the spreader bar, for supporting a patient harness or sling descends from the hoist assembly on a strap or a cable. The strap or cable is wound around a motorized drum for raising and lowering the patient harness or sling.
[0005] For example, the lift might be wheeled to position the hoist assembly and lifting member over or adjacent to a patient. The lifting member may then be lowered to receive the patient and subsequently raise the lifting member and patient so that they may be wheeled elsewhere to be lowered and placed. A ceiling lift may be utilized in a similar manner, however the hoist assembly is movably engaged to ceiling-mounted tracks such that the hoist assembly can be moved about the track from location to location.
[0006] A ceiling lift may be described as a motor unit movable along a rail, a flexible member is attached to a spreader bar. The motor unit commonly comprises a transmission, batteries and a control module.
[0007] The transmission is subjected to a number of challenges. For example, the transmission needs to be able to lift a patient, maintain the patient at a prescribed height for a certain period of time and lower the patient. Further, the transmission needs to be able to lift and support a weight of around 450 Kg.
[0008] In order to support such large weights, large motors capable of providing a high amount of torque are often used. Such motors may handle even heavy loads. However, large motors are usually expensive and consumes a lot of power.
[0009] To save cost and power, some manufacturers uses smaller motors able to deliver a high RPM. In order for the smaller motors to be able to support and lift higher loads, the RPM is often reduced and torque increased by means of different types of transmissions.
[0010] Such transmission systems are often complex and space consuming. Furthermore, due to the reliance on a fix transmission system, the motor unit may lack flexibility in terms of it being adaptable to different application, i.e. different types of patient lifts. In the light of the above, there

is a need for a transmission system which is associated with a low cost and high efficiency as well as flexibility.

SUMMARY

[0011] According to one aspect a drive system is provided. The drive system is for a patient lift. The drive system comprises a drum configured to control the vertical movement of a patient support mounting device of the patient lift via a load bearing member. The drive system further comprises at least one motor adapted to drive the drum, each motor being connected to a motor shaft gear via an output motor shaft.

[0012] Also, the drive system comprises a transmission connecting the motor and the drum. The transmission is adapted to transfer torque from the motor to the drum.

[0013] The transmission comprises a transmission interface adapted to interplay with the motor shaft gear. The transmission interface is configured to receive the motor shaft gear in at least two configurations. Each configuration is associated with an orientation of the output motor shaft relative the transmission interface.

[0014] According to an aspect, a patient lift is provided. The patient lift comprises the drive system, a patient support mounting device and a load bearing member. The patient support mounting device is connected to the drive system via the load bearing member.

[0015] According to an aspect, a method is provided. The method is for controlling a torque exerted by each of at least two motors comprised in a drive system. The drive system is configured to control the vertical movement of a patient support mounting device.

[0016] The method comprises obtaining a torque exerted by each of the motors, determining at least one torque differential value as a difference between the torque exerted by each of the motors and adjusting the torque exerted by at least one of the motors to compensate for the determined at least one torque differential value.

[0017] According to an aspect, a computer program product is provided. The computer program product is configured to, when executed by a control module, perform the method for controlling a torque exerted by each of at least two motors.

[0018] Further objects and features of the present invention will appear from the following detailed description of embodiments of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] The invention will be described with reference to the accompanying drawings, in which:

[0020] FIG. **1***a* is a perspective view of elements of a patient lift system.

[0021] FIG. **1***b* is a perspective view of elements of a patient lift system.

[0022] FIG. **2** depicts a drive system according to an embodiment implemented in a patient lift system.

[0023] FIG. **3** is a partial longitudinal-section view of a drive system according to an embodiment.

[0024] FIG. **4** is an exploded view of a drive system according to an embodiment.

[0025] FIG. **5***a* is a perspective view of a drive system according to an embodiment.

[0026] FIG. **5***b* is a perspective view of a drive system according to an embodiment.

[0027] FIG. **5***c* is a perspective view of a drive system according to an embodiment.

[0028] FIG. **6***a* is a perspective view of a drive system according to an embodiment.

[0029] FIG. **6***b* is a perspective view of a drive system according to an embodiment.

[0030] FIG. **6***c* is a perspective view of a drive system according to an embodiment.

[0031] FIG. **7** is a schematic view of the drive system according to an embodiment.

[0032] FIG. **8** is a perspective view of a locking arrangement and a motor of the drive system according to an embodiment.

[0033] FIG. **9** is a block diagram of a drive system according to embodiments of the invention.
[0034] FIG. **10** is a schematic flow chart of a method for controlling a torque exerted by each of at least two motors according to embodiments of the invention.
[0035] FIGS. **11***a* is a time plot of a pulse width modulation provided to a motor according to embodiments of the invention.
[0036] FIGS. **11***b* is a time plot of exerted torque by two motors according to embodiments of the invention.

DETAILED DESCRIPTION

[0037] FIGS. **1***a* and **1***b* show a non-limiting example of elements of a patient handling system with a patient lift. The patient lift may be in the form of a patient ceiling lift. A patient support mounting device **11** is connected via a load bearing member **12** to a lifting device **13** in FIG. **1***a*. The lifting device **13** may be arranged to be moveable along a track **14**. The lifting device **13** may thus be in engagement with the track **14**, e.g. movably connected to the track **14**. The lifting device **13** may move along the track **14**, preferably in both directions. The lifting device **13** may be in the form of a trolley movable along said track **14**.
[0038] The patient lift may comprise a drive system, which will be further described with reference to FIGS. **2-7**. The lifting device may comprise a drum for winding of the load bearing member **12** and motor and transmission for driving said drum. The load bearing member **12** may be wrapped around said drum for lowering and raising the patient support mounting device **11**. The drive system may be comprised in the lifting device.
[0039] In one embodiment, the lifting device **13** comprises wheels for interfacing with the track **14**. In one embodiment, the lifting device **13** is slidably connected to the track **14**.
[0040] The patient support mounting device **11** may be a spreader bar or hanger bar. The load bearing member **12** may be a flexible member such as a strap. The patient support **15** may, as shown in FIG. **1***b*, be a sling. The patient support mounting device **11** may be connected to the load bearing member **12** by means of a connecting unit **26**. The connecting unit **26** may be a quick connector, i.e. a connecting unit **26** adapted to receive the load bearing member **12** in a releasable manner.
[0041] The patient support mounting device **11** may comprise attachment elements **19** for attaching the patient support **15** to the patient support mounting device **11**. The attachment elements may comprise hooks with latches.
[0042] The lifting device **13** is configured to move the patient support mounting device **11** between a raised position situated closer to said lifting device **13** and a lowered position located more distantly from said lifting device **13**. The lifting device **13** may thus be configured to move the patient support mounting device **11** vertically between said raised and lowered position.
[0043] Although the patient lift in FIGS. **1***a*-**1***b* is depicted as a ceiling patient lift, the patient lift may also be floor lift with a base comprising a set of wheels for moving the lift across a floor.
[0044] FIGS. **2-7** discloses aspects of embodiments of a drive system for implementation in the patient lift depicted in FIG. **1**.
[0045] FIG. **2-4** depicts a drive system according to an embodiment. The drive system **100** comprises the drum and a transmission which may be comprised inside a housing **213′**. The drive system **100** thus comprises the casing **213′**. The drive system **100** further comprises at least one motor **270**. FIG. **3** discloses the drive system **100** without a part of the casing **213′**. The transmission and a part of an output motor shaft **274** is arranged inside the casing **213′**. The drive system comprises a locking arrangement **200** which will be further described with reference to FIG. **8**.
[0046] FIG. **4** discloses an explosion view of the drive system. The drive system comprises the drum **321**. The drum is configured to control the vertical movement of the patient support mounting device **11** via the load bearing member **12**. The patient support mounting device **11** is connected to the lifting device **13**, **213** via said load bearing member **12**. As previously described the drum **321**

is adapted to wind and unwind the load bearing member **12**. Thus, the drum is adapted to be connected to the load bearing member **12**. The load bearing member may be a flexible member such as a wire, cable or rope.

[0047] The drive system **100** further comprises the at least one motor **270**. The at least one motor **270** is adapted to drive the drum **321**. As depicted, the motor may be arranged orthogonally to the drum **321**. Each of the at least one motor **270** is connected to a motor shaft gear **227** via an output motor shaft **274**.

[0048] The output motor shaft **274** is arranged between the motor shaft gear **227** and the motor **270**. In one embodiment, the motor shaft gear **227** is connected via to the motor **270** by means of an additional gearing. In one embodiment, the motor shaft gear **227** may be directly connected to the motor **270**. Thus, the motor **270** may be directly connected to the input motor shaft, said input motor shaft comprising a gear, hence forming the motor shaft gear **227**. In one embodiment, the motor shaft gear **227** may be connected to an input shaft directly connected to the motor.

[0049] The drive system comprises the transmission **228**. The transmission **228** connects the motor **270** and the drum **321**. Thus, the motor **270** and the drum **321** are connected by means of the transmission **228**. The transmission **228** is adapted to transfer torque from the motor **270** to the drum **321**.

[0050] The transmission **228** comprises a transmission interface **220**. The transmission interface **220** is adapted to interplay with the motor shaft gear **227**. In other words, the transmission interface is adapted to interplay with the motor shaft gear **227** such that the drum **321** is driven by the motor **270**.

[0051] The transmission **228** comprises a transmission interface **220**. The transmission interface **220** is adapted to interplay with the motor shaft gear **227**. The transmission interface **220** is configured to receive the motor shaft gear **227** in at least two configurations. Each configuration is associated with an orientation of the output motor shaft **274** relative the transmission interface **220**.

[0052] In the field of patient lifts, the requirements on the drive system may vary greatly depending on the application of the patient lift and the weight and mobility of the patient to be carried by the patient lift. The transmission potentially being able to receive torque from the at least one motor in more than one manner, i.e. configuration enables a modular solution where more than one motor may be utilised or the positioning of the motor may be altered depending on the available space. Thus, a drive system which allows for an increased flexibility in terms of usage is achieved.

[0053] A configuration is herein defined as a position in which the motor shaft gear **227** interfaces with the transmission interface **220**. Consequently, the position of the motor shaft gear **227** relative the transmission interface **220** is provided by means of a corresponding orientation (i.e. direction and position) of the output motor shaft **274** relative the transmission interface **220**.

[0054] The transmission interface **220** thus is adapted to be in direct engagement with the motor shaft gear **227**, i.e. adapted to be directly connected to the motor shaft gear **227**.

[0055] The transmission interface **220** may be in the form of one or multiple gears or a belt drive wheel etc.

[0056] In one embodiment, the transmission interface **220** comprises an input transmission gear. The input transmission gear **323** is adapted to interplay with the motor shaft gear **227**.

[0057] In one embodiment, wherein the drive system **100** comprises more than one motor **270**, the input transmission gear **323** is adapted to interplay with a first motor shaft gear **227** connected to the first motor and a second motor shaft gear **227** connected to the second motor. This allows for a drive system which is simple to install and is space efficient as well as less complex compared to other modular drive systems for patient lifts.

[0058] In another embodiment, the transmission interface **220** may comprise a plurality of input transmission gears **323**. Thus, a first input transmission gear **323** may be adapted to interplay with the first motor shaft gear **227** connected to the first motor **270**. A second input transmission gear **323** may be adapted to interplay with the second motor shaft gear **227** connected to the second motor **270**.

[0059] In one embodiment, the input transmission gear **227** may be arranged orthogonally to the drum **321**. Thus, the engaging portion of the input transmission gear **227** may extend orthogonally to the drum **321**. As depicted in FIG. **4**, the input transmission gear may be a cogged wheel. The periphery of the cogged wheel may thus extend orthogonally to the drum **321**.

[0060] Further referencing FIG. **4**, the motor shaft gear **227** and the input transmission gear may form a worm drive. As is well-known for the skilled person, a worm drive is formed by a worm gear and a worm wheel. The worm wheel is arranged orthogonal to the worm gear.

[0061] In one embodiment, the motor shaft gear **227** may be a worm gear. The input transmission gear **323** may thus be a worm wheel. In one embodiment, motor shaft gear **227** may be arranged orthogonal to the input transmission gear **323**. This allows for the input transmission gear **323** to receive the motor shaft gear **227** in different configuration in a space efficient and non-complex manner.

[0062] In an alternative embodiment, the motor shaft gear **227** may be a worm wheel and the input transmission gear **323** may thus be a worm gear.

[0063] The transmission **228** may further comprise an output gear **322**. The output gear **322** is fix to the drum **321**. The output gear **322** is connected to the transmission interface **220** for receiving torque from the motor shaft hear **227**. Thus, the output gear is arranged between the transmission interface **220** and the drum **321** for transferring torque between said transmission interface **220** and drum **321**.

[0064] The output gear **322** may comprise a ring wheel. The ring wheel is fix to the drum **321**. Thus a more compact drive system is achieved.

[0065] The output gear **322** may be coaxial with the drum **321**.

[0066] In one embodiment, the ring wheel may be an integrated part of the drum **321**. In one embodiment, the transmission **228** may comprise a planetary gear wheel **326**. The planetary gear wheel **326** interfaces with the ring wheel **322**. The planetary gear wheel **326** is arranged between the transmission interface **220** and the ring wheel **322**.

[0067] FIG. **5***a*-*c* depicts various embodiments of the drive system. As seen in the figures the transmission interface **220** is configured to receive the motor shaft gear **227** in at least two configurations. Each configuration is associated with an orientation of the output motor shaft **274** relative the transmission interface **220**.

[0068] FIG. **5***a* depicts a drive system wherein transmission interface **220** receives the motor shaft gear in one of the configurations of the at least two configurations. The output motor shaft **274** may have an orientation which is orthogonal to the drum **321** in the at least two configurations. As seen in said FIG. **5***a*, the orientation of the output motor shaft **274** associated with the depicted configuration of the transmission interface and motor shaft gear is substantially orthogonal to the drum.

[0069] In an alternative embodiment, the orientation of the output motor shaft **274** may have any other orientation relative the transmission interface **220**, however such a solution requires additional gearings and is less beneficial.

[0070] FIG. **5***b* depicts a drive system wherein the transmission interface **220** receives the motor shaft gear in another one of the at least two configurations. As seen in said FIG. **5***b*, the output motor shaft **274** is oriented orthogonal in one of the configurations relative to output motor shaft **274** in another one of the configuration. Thus, a first configuration depicted in FIG. **5***a* is associated with an orientation of the output motor shaft which is orthogonal relative the orientation of the output motor shaft, the second configuration being associated with said orientation of the output motor shaft.

[0071] FIG. **5***c* depicts a drive system comprising two motors. The input motor gear shaft **227** of the first motor has a first orientation and the input motor gear shaft **227** of the second motor has a second orientation. The orientation of the input motor gear shafts are substantially parallel. Thus, the output motor shaft **274** is oriented parallel in one of the configurations relative to the output motor shaft **274** in another one of the configurations. Accordingly, the transmission interface **220** is configured to receive the motor shaft gear **227** in at least two configurations. One of the at least two configurations is associated with an orientation of the output motor shaft **274**. The other one of the at least two configurations is associated with an orientation of the output motor shaft **274**, said orientation being parallel to the orientation of the output motor shaft in the first output motor shaft **274**. Albeit the drive system of FIG. **5***c* is illustrated as comprising two locking arrangements **200**, it should be mentioned that this is but one embodiment, and a drive system comprising two motors **270** may very well be formed without any, or only one, locking arrangement **200**. For example, only one motor **270** may be provided with a locking arrangement **200**. A second motor **270** may instead be provided with spacers to replace said locking arrangement **200**. A single locking arrangement **200** provided on one of the motors may thus be utilized to brake a drive system comprising a plurality of motors by means of locking the motor shaft gear connected to said one of the motors.

[0072] The drive system may comprise a first and second motor **270**. The transmission interface **220** is thus adapted to interplay with a first motor shaft gear **227** connected to the first motor **270** and a second motor shaft gear **227** connected to the second motor **270**. The first motor shaft gear **227** is connected to the first motor **270** via the first output motor shaft **274**. The second motor shaft gear **227** is connected to the second motor **270** via the second output motor shaft **274**. The transmission interface **220** is adapted to interplay with the first motor shaft gear **227** connected to the first motor **270**. The transmission interface **220** is further adapted to interplay with the second motor shaft gear **227** connected to the second motor **270**.

[0073] Having two motors for driving and controlling the drum is associated with a number of advantages. It allows for usage of smaller motors instead of one larger to provide a high torque to the drum. Furthermore, having smaller motors allows usage of cheaper motors. Also, having two motors allows for a modular system where smaller electric components such as circuit boards may be used for multiple applications. Having singular large electronic motors requires larger electric components, which may not be suitable for every implementation.

[0074] In one embodiment, the first motor shaft gear **227** and the second motor shaft gear **227** are parallel. Thus, the transmission interface **220** receives the first and second motor shaft gear **227** in configurations associated with a first and second orientation of the first and second output motor shaft **274**, respectively. The first orientation being parallel to the second orientation. This allows for implementation of two motors in a space efficient manner.

[0075] In one embodiment, the first and second orientation may be parallel and opposite. Thus, the first output motor shaft may extend in a direction opposite to the second output motor shaft.

[0076] In one embodiment, the first and second orientation may be parallel and in the same direction. Thus, the first output motor shaft may extend in the same direction and parallel to the second output motor shaft.

[0077] The first motor **270** may be arranged at a first side relative the transmission interface **220** and the second motor **270** may be arranged at a second side relative the transmission interface **220**. The second side may be opposite to the first side. With reference to FIG. **6***a-c*, the motor **270** may have different sizes and capacity. Thus the transmission interface **220** may be adapted to interchangeably receiving the input motor gear shaft **227** connected to the motor **270**. This allows for a drive system which may be implemented in a wide array of patient lifts due to flexibility of the system. Hence, the motor **270**, the output motor shaft **274** and the motor shaft gear **227** may be arranged to form a motor module. The transmission interface **220** may be adapted to interchangeably receiving the motor module.

[0078] FIG. **7** schematically depicts the drive system according to an embodiment in further detail.

[0079] The motor shaft gear **227** interfaces with the transmission interface. The

text missing or illegible when filed

[0080] text missing or illegible when filed **20** transmission interface **220** comprises the input transmission gear **323**.
[0081] The transmission **228** may comprise a first gear **325** connected to the input transmission gear **323**. The first gear **325** may be coaxial to the input transmission gear. In one embodiment, the first gear **325** may be coaxial to the ring wheel **322**. In one embodiment, the first gear **325**, the input transmission gear **323** and the ring wheel **322** may be coaxial. The coaxial design of the transmission allows for a more compact transmission which enables sufficient torque transfer to the drum.
[0082] The transmission **228** may comprise an input transmission shaft **431**. The input transmission shaft **431** being arranged to transfer torque from the input transmission gear **323** to the first gear **325**. The first gear **325** and the input transmission gear **323** may both be mounted to the input transmission shaft **431**.
[0083] The first gear **325** may be connected to the ring wheel **322** via an intermediate gearing. The intermediate gearing is adapted to transfer torque from the first gear **325** to the ring wheel **322**.
[0084] In one embodiment, the intermediate gearing comprises a first intermediate gear **324**. The first intermediate gear **324** interfaces with the first gear **325**.
[0085] The intermediate gearing may further comprise a second intermediate gear **326**. The second intermediate gear **326** may be connected to the first intermediate gear **324**. The second intermediate gear **326** may be adapted to transfer torque from the first intermediate gear **324** to the ring wheel **322**. In one embodiment, the first and second intermediate gear may be coaxial. In one embodiment, the second intermediate gear **326** may interface with the ring wheel **322**.
[0086] In one embodiment, the intermediate gearing comprises an intermediate shaft **432**. The intermediate shaft **432** may be adapted to transfer torque from the first intermediate gear **324** to the second intermediate gear **326**. The first and second intermediate gear may be mounted to the intermediate shaft **432**.
[0087] Further referencing FIG. **7**, the brake element **329** may be fixedly mounted to the ring wheel **322** and/or the drum **321**. The brake clement **329** may be coaxial with the ring wheel **322**. In one embodiment, the brake element **329** may be coaxial with the any or all of the input transmission gear **323**, the first gear **325** and the intermediate shaft **431**.
[0088] In one embodiment, the transmission **228** may comprise a planetary gearing. Thus, the first gear **325** may be a sun gear of the planetary gearing. Further, the intermediate gearing may comprise a planet gear. In one embodiment, the first intermediate gear **324** is a planet gear interfacing with the sun gear, i.e. the first gear **325**.
[0089] In one embodiment, at least one motor **270** of the at least one motor **270** is provided with the locking arrangement **200**. The locking arrangement **200** is configured to selectively lock the motor shaft gear **227**.
[0090] In one embodiment, each motor **270** of the at least one motor **270** may be provided with the locking arrangement **200**. The locking arrangement **200** is configured to selectively lock the motor shaft gear **227**.
[0091] FIG. **8** depicts the locking arrangement in closer detail. The locking arrangement **200** may be configured to switch from a disengaged mode in which the locking arrangement **200** does not lock the motor shaft gear **227** to an engaged mode in which the locking arrangement **200** locks the motor shaft gear **227**. This may occur in response to the motor **270** switching from an operating state to a powerless state.
[0092] In one embodiment, the locking arrangement **200** may be configured to switch from the engaged mode to the disengaged mode in response to the motor **270** switching from the powerless state to the operating state.
[0093] In one embodiment, the locking arrangement may comprise a shape memory alloy element **251** and a locking device **250**. The shape memory alloy element is connected to said locking device **250** and arranged to selectively actuate said locking device **250** to control a locking force on an

engagement member **273**. The engagement member **273** is mechanically connected to the motor **270** and the load bearing member **12** of the patient lift, i.e. the motor **270** of the patient lift and the load bearing member **12** of the patient lift. Said motor **270** is arranged to raise and lower the patient support mounting device **11**.

[0094] In the engaged mode, the shape memory alloy element **251** is in a first configuration and the locking device **250** is in an engaged position for exerting a locking force on the engagement member **273** thereby preventing vertical movement of the patient support mounting device **11**.

[0095] In the disengaged mode, the shape memory alloy element **251** is in a second configuration actuating the locking device **250** to a disengaged position in relation to the engagement member **273** thereby enabling vertical movement of the patient support mounting device **11**.

[0096] Compared to known patient lifts implementing locking worm gear transmissions this allows for locking without creeping even when a large load is suspended by means of the patient support mounting device **11**. Furthermore, the shape memory alloy allows for a more cost-efficient and less power consuming solution compared to a solenoid activated mechanical brake. Further, this allows for the locking device and the motor to form a single module. Hence, the adaptability of the drive system is further enhanced due to both the motor and locking functionality being provided in the form of a module.

[0097] A shape-memory alloy is as is known in the prior art an alloy which can be deformed in a cold state but returns to a pre-deformed shape when heated. Shape-memory alloys are also known in the prior art as memory metals, memory alloys, smart metals, smart alloys or muscle wires.

[0098] The shape memory allow element **151**, **251** may be in one of: Ag—Cd, Au—Cd, Co—Ni—Al, Co—Ni—Ga, Cu—Al—Ni, Cu—Al—Ni, Cu—Al—Ni—Hf, Cu—Sn, Cu—Zn, Cu—Zn—Si, Cu—Zn—Al, Cu—Zn—Sn, Fe—Mn—Si, Fe—Pt, Mn—Cu, Ni—Fe—Ga, Ni—Ti, Ni—Ti—Hf, Ni—Ti—Pd, Ni—Mn—Ga, Ti—Nb alloy.

[0099] The shape memory alloy clement **251** may be a two-way memory effect element. In the first configuration, the shape memory clement **251** forms a shape which allows the locking device **250** to in the engaged position in relation to the engagement member **273**. In the second configuration **251** forms a shape which is arranged to force the locking device to the disengaged position in relation to the engagement member **273**.

[0100] The locking device **250** may thus be a movable by means of the shape memory alloy clement **251**. Accordingly, the shape memory alloy clement **251** may be arranged to move the locking device **250** between the engaged position and disengaged position The shape memory alloy clement **251** may be directly attached to the locking device **250**.

[0101] In one embodiment, the shape memory alloy clement **251** is a muscle wire.

[0102] The shape memory alloy element **251** may be arranged to be electrically connected to at least one power source **340** for selectively transitioning between the first and second configuration.

[0103] The locking arrangement is arranged to switch from the disengaged mode to the engaged mode in response to no power being provided to the motor **270**. The locking arrangement may thus function as an emergency brake which is actuated in response to the patient lift not being supplied with power. As soon as power is supplied to the motor **270** the locking arrangement switches from the engaged mode to the disengaged mode, which allows for normal operation of the patient lift.

[0104] According to an aspect, a patient lift is provided. The patient lift comprises the drive system according to any one of the previously described embodiments. The patient lift further comprises the patient support mounting device **11** and the load bearing member **12**. The patient support mounting device is connected to the drive system via the load bearing member.

[0105] With reference to FIG. **9**, a simplified block diagram of the drive system **100** is shown. In embodiments of the drive system **350**, the at least one motor **270** is controlled by a controller **350**. The controller **350** may be comprised in the drive system **100** or be provided as an external control module **350**. The control module may be any suitable controller and the invention is not limited by specifics regarding the control module **350**. The control module **350** will typically be operatively connected to the power source **340** and the motor(s) **270**. It should be mentioned that also the power source **340** may be comprised in in the drive system **100** or external to the drive system **100**. The power source **340** may be any suitable power source **340** and the skilled person will know how to implement and/or adapt the disclosed invention to function with direct current, DC, alternating current, AC, current sources or voltage sources of any level. The control module **350** may further be operatively connected to any or all other parts of the drive system **100** in order to obtain torque readings from e.g. the drum **321**. The control module **350** may further be operatively connected to a user interface for controlling the patient support mounting device **11**. The control module **350** may be realized using a single control system or may be implemented using a distribute system with sensors and/or controllers distributed throughout the drive system **100** and/or the patient lift. The term operatively connected is to mean any suitable connection and may be direct connection, a wired connection, a wireless connection, a connection via a BUS or a connection via active circuitry or logic.

[0106] The inventors behind this disclosure have further realized that issues may arise when controlling more than one motor **270** driving a common drum **321** as can be the case in the disclosed drive system **100**. If all motors are not transferring substantially the same amount of torque to the drum **321**, the motor **270** providing the most torque may actually drive any other motor **270** in the drive system **100**. Consequently, the torque contributed by each motor **270** should be approximately the same for all motors **270** in the drive system unless mechanical complexity is to be added in the transferral of torque from each motor **270**.

[0107] Typically, the motors **270** of drive system **100** are controlled by a current provided to them from a power source **340**. The simplest way of controlling the motors **270** is to use the same controlled current for all motors **270**. A preferred alternative is to control each of the motors **270** individually in order to allow e.g. current and safety limitations to be applied to each motor **270**. On the other hand, having more than one motor **270** driving a common drum **321** may introduce problems as the motors **270** may contribute differently to the drive of the drum **321**. One motor **270** may exert almost all torque that drives the drum **321** and the other(s) may be virtually idle when it comes to contribution of torque. This may cause added wear to the motor **270** contributing most to the drive of the drum **321**. In this case, it is also preferred to control each of the motors **270** individually.

[0108] When each motor **270** is controlled individually each motor **270** is provided with an input power P.sub.in that can calculated as the product of a voltage V.sub.in and a current I.sub.in provided to the motor **270**. The power out P.sub.out from the motor **270** can be described as the torque T provided by the motor **270** multiplied by the speed, Revolutions Per Minute, RPM, the revolutions of the motor **270**. Since the motors **270** of the drive system **100** are joined together, they all have the same speed. Consequently, assuming the same efficiency of all motors **270**, any difference in input power P.sub.in between the motors **270** can be attributed to a difference between the motors **270** in the torque they provide to the drum **321**.

[0109] In order to mitigate these problems, a method **400** for controlling the torque exerted by each of at least two motors **270** comprised in a drive system **100** will be described with reference to FIGS. **9-11**. The method **400** may be run on top of, in addition to, or as an extension to another motor control method e.g. a method for soft start, controlled braking etc. The conceptual idea of the method **400** is to ensure that the effort of driving the drum **321** is shared substantially equal between all motors **270** of the drive system **100**. This will increase the life-time of the motors **270** and the drive system **100** since e.g. not one motor shaft gear **227** is subjected to more stress than the other motor shaft gears **227**. Naturally, this reason applies to all parts of the drive system **100**.

[0110] In order to equalize the torque provided by each motor **270**, the torque exerted by each motor **270** is acquired **410**. The torque may be acquired **410** directly by e.g. using a Newton meter, however, such instrumentation is costly and increases the cost of the motor **270** and/or the drive system **100**. An alternative, and preferred way of acquiring **410** the torque is to estimate it based on the current provided to the motor **270**. In many cases the current I.sub.in provided to the motor is controlled by Pulse Width Modulation, PWM, of a power source **340**. From hereon, the term PWM will typically mean the duty cycle of the PWM although not specifically stated, this will be obvious to the skilled person. The power source **340** is typically a voltage source supplying a voltage V.sub.in that is effectively reduced by the PWM such that the input power P.sub.in of the inductive load of the motor **270** can be accurately controlled. Since the speed of all motors **270** is the same, the inventors have realized that a metric proportional to the torque of the motor **270** may be acquired **410** by fractioning the average current provided to the motor **270** by a duty cycle of the PWM. Hereinafter, changing, adjusting or otherwise adapting the PWM, is to mean changing the duty cycle of the PWM. Methods for measuring and averaging the input current I.sub.in is known to the skilled person and both analogue, e.g. low pass filtering, or digital averaging of the current may be used. In a drive system with N motors **270**, the average current provided to the respective motors **270** is denoted I.sub.n, and the duty cycle of the corresponding PWM is denoted PWM.sub.n. Each of the currents I.sub.n is divided by the associated PWM.sub.n to a torque metric T.sub.n as shown in Eqn. 1 below.

T.sub.n=*I*.sub.n/PWM.sub.n Eqn. 1

[0111] A torque error e.sub.n,m can be determined **420** as the difference between a motor n and another motor m according to Eqn. 2

e.sub.n,m=*T*.sub.n−*T*.sub.m=*I*.sub.n/PWM.sub.n−*I*.sub.m/PWM.sub.m Eqn. 2 [0112] Wherein n and m reference specific motors **270** of the n motors **270**. n can be any number between 1 and ∞, i.e. an arbitrary number, and consequently n and m can be any number between 1 and n.

[0113] In other words, if the drive system **100** comprises three motors **270**, two torque errors e.sub.n,m will typically be calculated for each motor **270**, that is e.sub.1,2, e.sub.1,3, e.sub.2,1, e.sub.2,3, e.sub.3,1 and e.sub.3,2.

[0114] In one embodiment, n in the Eqn. 2 above, always refer to the motor with the weakest torque, i.e. T.sub.n≤T.sub.m. In this embodiment, the motor **270** contributing the least torque to the drum **321** will be regarded as the master and other motors **270** as slaves. The torque of the master is the torque that the other motors **270**, the slaves, will use as target torque when controlling the torque, as will be detailed in coming sections. In this embodiment, torque errors need only be determined with reference to the weakest torque T.sub.n. In order to exemplify, in the drive system with three motors **270**, assume that motor #1 is contributing the least torque to the drum **321**. This means that, in this embodiment, only e.sub.1,2, e.sub.1,3 torque errors are necessary to calculate. Note that the motor **270** determined to be the master can change during the control of the if, for instance, for one of the slaves, the PWM is at the maximum and the torque is lower than the master's torque.

[0115] The torque error may also be referenced as a torque differential value.

[0116] From the torque errors e.sub.n,m, it is possible to determine how each motor **270** contributes to the drive of the drum **321**. Different control strategies may be employed, either the motor(s) **270** contributing the most torque will have their torque decreased, or the motor(s) **270** contributing the least torque will have their torque increased. Alternatively, the strategies may be combined and the motor(s) **270** contributing the most torque will have their torque decreased and the motor(s) **270** contributing the least torque will have their torque increased such that the torque of each motor converges on an intermediate torque. Different control strategies may be employed depending on the use case. If for instance the drum **321** is in the process of lowering a patient, there would typically be a speed limitation that must not be exceeded, and this is typically linked to an upper limit in the PWM duty cycle. Once one of the motors **270** reaches this PWM limit, the other motor(s) are controlled such that they provide the same torque or reach the PWM limit. If the PWM limit is reached by the other motors without the torque being the same, the motor **270** first reaching the PWM limit is controlled to reduce its torque until it is substantially the same as the other motors.

[0117] To clarify the need for control, further explanation will be provided with reference to FIGS. **11***a-b*. This explanation is given with two motors **270**, but the skilled person will, after reading this disclosure, be able to expand the teachings to control more than two motors **270**. The motors **270** are assumed to be of the same model and delivered according to a common specification. The drive system **100** is controlled to rotate the drum **321** such that a load, e.g. a patient, is lifted. The acceleration is controlled and is to a linear path until the desired speed is achieved at which point the acceleration stops and the speed is kept constant. The speed may be controlled by having a target PWM that corresponds to the desired speed. FIG. **11***a* illustrates how the PWM may change over time as the drum is accelerated for a first period of time A after which the speed is constant during a second period of time S until it is finally de-accelerated during a third period of time D. The PWM as illustrated in FIG. **11***a* is applied to both motors **270** and in FIG. **11***b*, the torques, T**1**, T**2**, exerted by each of the motors **270** is illustrated. The motor **270** exerting the torque T**1**, dotted line in FIG. **11***b*, is illustrated as exerting a lower torque than the motor **270** exerting the torque T**2**, solid line in FIG. **11***b*. These torques T**1**, T**2** are, as taught in Eqn. 1, proportional to their respective currents and PWM's. Since the same PWM is supplied to both motors **270**, in this example, the currents provided to each motor **270** would exhibit a behaviour similar to that of the torques T**1**, T**2** illustrated in FIG. **11***b*. The reason for the currents, and consequently the torques, being different may be e.g. aging, malfunction, individual differences etc. As mentioned earlier, since both motors **270** are operating at the same speed, the differences seen in FIG. **11***b* result in the first motor **270**, contributing less torque to the drum **321** and added wear to the second motor **270** as a result. With continued reference to FIG. **11***b*, if instead each motor **270** is controlled by an individual PWM, reducing the PWM of the second motor **270** would decrease the second torque T**2**, increasing the PWM of the first motor would increase the first torque T**1**. Consequently, by controlling the PWM based on the torque, or rather the torque error en,m as explained with reference to Eqn. 2, it is possible to change the PWM such that all motors **270** contribute substantially the same torque to the drum **321**.

[0118] Returning to the method **400** and FIG. **10**, the determined **420** torque error is, as explained in previous sections, used to adjust **430** the torque exerted by at least one of the motors **270**. The torque may, as is understood from the previous sections, be controlled by adjusting the PWM. The adjustment **430** may be accomplished by an Adjusted Power Level, APL, that is applied to the PWM associated with the motor **270** to be controlled. The APL is used as a factor on the PWM and the APL may be restricted depending on the control strategy, e.g. if no increase is allowed, the APL may be limited with 1.0 as its maximum value and if no decrease is allowed, the APL may be limited with 1.0 as its minimum value. Preferably, there will be one APL associated with each motor **270** of the drive system **100**. In this disclosure, APL of 1.0 will typically correspond to no compensation and an APL lower than 1.0 correspond to a decrease of the PWM and an APL greater than 1.0 will correspond to an increase of the PWM. This is not to be considered a limiting factor and the skilled person realises that by, for instance, dividing the PWM with the APL, the reverse association will be achieved. As a starting value, the APL is preferably 1.0 and is then compensated based on the torque error en,m. The APL may be updated by simply subtracting the associated torque error en,m from the current APL, but preferably the torque error e.sub.n,m is processed by a P, PI, PD or PID controller which are all known from the art.

[0119] Returning to FIG. **10** and the method **400** for controlling the torque exerted by each of at least two motors **270** comprised in the drive system **100**. The method **400** may in embodiment be run continuously or a predefined or configurable number of times. The method **400** may be initiated by e.g. operation of the drive system **100** or upon detection of movement of one of the motors **270**. As mentioned a torque exerted by each of the motors **270** is acquired **410**. The acquired torque is used to determine **420** torque error(s) as a difference between the torque exerted by a first motor **270** of said at least two motors **270** and the torque exerted by each of the other at least two motors **270**. This may be done as described above with reference to Eqn. 1 and 2. Based on the determined torque error, the torque exerted by at least one of the motors **270** is adjusted **430**. The torque is adjusted in order to compensate for the determined **420** torque error. Depending on how the method **400** is implemented, the entire error may be compensated, but preferably, a controller e.g. a P, PI, PD or PID controller, is utilized in order to smoothly compensate for the torque error over a number of iterations of the method **400**.

[0120] In one embodiment of the method, it further comprises, after, or as part of, the step of determining **420**, a step of updating **425** the previously disclosed APL for at least one of the motors **270** of the drive system **100**. In a preferred embodiment of the method **400** executed on a drive system **100** comprising two or more motors **270**, the APL is updated for each of these motors **270**.

[0121] In a further optional embodiment, the step of adjusting **430** is performed by scaling the torque exerted by at least one of the motors **270** with the APL associated with said at least one of the motors **270**.

[0122] In an optional embodiment of the method **400**, the APL of each motor is limited to a maximum value of 1.0. From this follows that the torque of the motor **270** contributing the least torque to the drum **321** will be used as a target torque, i.e. motor(s) **270** contributing more torque will be associated with an APL<1.0 and consequently have their PWM and torque contributed reduced. This means determining which motor **270** contributes the least torque and reducing the torque contributed by the other motors **270** to substantially the same torque level as that of the motor **270** contributing the least torque.

[0123] In another optional embodiment of the method **400**, a speed limit and/or speed target is applied to the drive system **100**. The speed limit and/or speed target is typically associated with a resulting rotational speed of the drum **321** but may be any speed affected by the motors **270**. In this embodiment, the method **400** further comprises, determining **427** a target current and/or a target PWM associated with the speed limit and/or speed target. This may be achieved through e.g. a predefined or configurable equation or look up table.

[0124] In a further optional embodiment, each of the motors **270** is controlled **429** based on the determined **427** target current and/or target PWM, until one of the motors **270** reaches the target current and/or the target PWM. When one of the motors **270** reaches the target current and/or the target PWM, the step of adjusting **430** is applied to only to the other motors **270**, i.e. the motors of the drive system **100** not having reached the target current and/or the target PWM. The steps of determining **427** the target and controlling **429** the motors may be run integrated with the method **400** or in parallel with the method **400**.

[0125] Alternatively, when controlling the speed, not all of the motors **270** are controlled to reach the determined **427** target current and/or target PWM. Any motors **270** not being targeted to reach the determined **427** target current and/or target PWM may effectively have a braking effect on the drum and act as generators (depending on the chosen type of motor **270**). This may be achieved by e.g. not applying a PWM or current, or applying a PWM or current that is lower than the target current/PWM, to motors not being targeted to reach the determined **427** target current and/or target PWM.

[0126] In one optional embodiment of the method **400**, no adjusting **430** for difference in torque is until the PWM for each of the motors is above 10%, preferably above 20% and most preferably above 25%. This is beneficial since the measured average current is divided by the PWM, any measurement error of the current will impact the calculated torque error e.sub.n,m more for lower PWM duty cycles.

[0127] In one optional embodiment of the method **400**, the control of the APL is slow. This may mean that the APL or the torque error e.sub.n,m is averaged over a time period that is an accumulated time period of operation of the drive system **100**. In this context, operation of the drive system **100** is to mean operation of at least one of the motors **270**, i.e. providing a PWM with

a duty cycle larger than 0 to at least one of the motors. It may be that the accumulated time period of operation is only accumulated when e.g. the PWM is above or below a PWM threshold or when the PWM is substantially constant, i.e. no acceleration of the drum **321**. In a further embodiment of the method **400**, the torque error is averaged over an accumulated time period of operation of the drive system **100** that is longer than 30 s, preferably longer than 60 s and most preferably longer than 120 s. In an even further embodiment, the accumulated time period of operation is only accumulated when the PWM is above 10%, preferably above 20% and most preferably above 25%.
[0128] In one optional embodiment of the method **400**, the APL associated with each motor **270** is stored in a persistent manner such that it may be retrieved again after e.g. a power failure. In an alternative embodiment of the method **400**, the APL associated with each motor is reset to 1.0 each time power is lost.
[0129] The method **400** may be altered, adjusted or tuned in numerous ways and the presentation above is supposed to give a general idea of the concept and is not intended to detail all thinkable variants. The embodiments presented above may be combined in any suitable way. After reading this disclosure, the skilled person will realize that for instance the APL can be limited to 1.0 such that only decrease of PWM is allowed. One of the motors **270** may be selected as a master and the other motor(s) will be controlled to adjust their respective torque to be as close as possible to the torque of the master.
[0130] The method **400** may be executed by any suitable electric circuitry or performed by a suitable controller executing software code implementing the method **400**.
[0131] The described torque error e.sub.n,m or the presented APL may be of further use, other than ensuring that all motors **270** are contributing equally to the torque of the drum **321**. If the APL is far from 1.0, this may be a sign of malfunction or ware of the system. The term far from 1.0 is vague and the skilled person will know, after reading this disclosure, what difference, error e.sub.n,m or APL is to be considered significant in determining the health of the system. It may be that a 10% deviation from 1.0 in the APL is significant in one system, and a 25% deviation is significant in another system. The drive system **100** may be configured to act upon a significant difference in APL or error e.sub.n,m. A limit for acting may be predetermined or configurable and the action taken may be any suitable action e.g. generating an alert or stopping the drive system **100**. The drive system **100** may further be configured to track, collect and/or log data pertaining to the exerted torque, the error e.sub.n,m, the APL and/or any other parameter in the drive system **100** such that statistical analysis may be performed on the data.
[0132] According to an aspect, a computer program is provided. The computer program product is configured to, when executed by a control module, perform the method for controlling a torque exerted by each of at least two motors of any of the above embodiments.
Clauses
[0133] The scope of the invention is defined in the appended claims and the following clauses are to be considered exemplary embodiments of the invention.
[0134] Clause 1. A drive system (**100**) for a patient lift, the drive system comprising: [0135] a drum (**321**) configured to control the vertical movement of a patient support mounting device (**11**) of the patient lift via a load bearing member (**12**), [0136] at least one motor (**270**) adapted to drive the drum (**321**), each motor (**270**) being connected to an motor shaft gear (**227**), [0137] a control module (**350**) operatively connected to said at least one motor (**270**) and a power source (**340**), and [0138] a transmission (**228**) connecting the motor (**270**) and the drum (**321**), the transmission (**228**) being adapted to transfer torque from the motor (**270**) to the drum (**321**), [0139] whereby the transmission (**228**) comprises a transmission interface (**220**) adapted to interplay with the motor shaft gear (**227**).
[0140] Clause 2. The drive system (**100**) of Clause 1 wherein the transmission interface (**220**) is configured to receive the motor shaft gear (**227**) in at least two configurations, each configuration being associated with an orientation of the output motor shaft (**274**) relative the transmission

interface (**220**).

[0141] Clause 3. The drive system (**100**) of Clause 1 or 2, wherein the control module (**350**) is configured to control a torque exerted by said at least one motor (**270**) by controlling a power supplied to said at least one motor (**270**) from the power source (**340**).

[0142] Clause 4. The drive system (**100**) of Clause 33 wherein the controller is further configured to obtain the torque exerted by said at least one motor (**270**) based on an average current and a Pulse Width Modulation, PWM, duty cycle setting provided to said at least one motor (**270**).

[0143] Clause 5. The drive system (**100**) of Clause 3 or Clause 44, wherein the control module (**350**) is configured to control the power supplied to said at least one motor (**270**) from the power source (**340**) substantially continuously.

[0144] Clause 6. The drive system (**100**) of Clause 5, wherein the control module (**350**) is further configured control the power supplied to said at least one motor (**270**) based on a control parameter comprising a product part.

[0145] Clause 7. The drive system of Clause 6, wherein the control parameter further comprises an integral part.

[0146] Clause 8. The drive system (**100**) of Clause 6 or Clause 77, wherein the control parameter further comprises an derivative part.

[0147] Clause 9. The drive system (**100**) of any of Clause 4 to Clause 8, wherein a speed limit is applied to the drive system (**100**), and the control module (**350**) is further configured to: [0148] determine a target current and/or a target PWM duty cycle associated with the speed limit, and [0149] control said at least one motor (**270**) until at least one motor (**270**) reaches the target current and/or the target PWM duty.

[0150] Clause 10. The drive system (**100**) according to Clause 9, wherein only one of said at least one motor (**270**) is controlled until it reaches the target current and/or the target PWM duty.

[0151] Clause 11. The drive system (**100**) of any of the preceding Clauses, comprising at least two motors (**270**), wherein the shaft gears (**227**) associated with each of said at least two motors (**270**) are rotating at substantially the same number of Revolutions Per Minute, RPM.

[0152] Clause 12. The drive system (**100**) of Clause 10, wherein the control module (**350**) is further configured to: [0153] obtain the torque exerted by each of said at least two motors (**270**), [0154] determine at least one torque differential value as a difference between the torque exerted by each of said at least two motors (**270**), and [0155] adjusting the torque exerted by at least one of said at least two motors (**270**) to compensate for the determined at least one torque differential value.

[0156] Clause 13. The drive system (**100**) of Clause 11, wherein the control module (**350**) is further configured to, before determining said least one torque differential value, update an Adjusted Power Level, APL, for each of said at least two motors (**270**).

[0157] Clause 14. The drive system (**100**) of Clause 12, wherein the control module (**350**) is configured to adjust the torque exerted by at least one of said at least two motors (**270**) by scaling the torque exerted by at least one of the motors (**270**) with the APL associated with said at least one of the motors (**270**).

[0158] Clause 15. The drive system (**100**) of any of Clause 10 to Clause 13, wherein the control module (**350**) is further configured to, when said at least one motor (**270**) reaches the target current and/or a target PWM duty cycle, adjust the torque exerted by all motors (**270**) except said at least one motor (**270**) first reaching the target current and/or the target PWM duty cycle.

[0159] Clause 16. The drive system (**100**) of any of Clause 10 to Clause 14, wherein the control module (**350**) is further configured determine which motor (**270**) contributes the least torque and adjust reduce the torque exerted by each of the other motors (**270**) to substantially the same torque as the torque contributed by the motor (**270**) contributing the least torque.

[0160] Clause 17. The drive system (**100**) of any of Clause 1010 to Clause 16, wherein the torque exerted by each of the motors (**270**) is controlled by the control module (**350**) based on at least a PWM duty cycle, and the control module (**350**) is further configured to start adjusting the torque

exerted by at least one of said at least two motors (**270**) when the PWM duty cycle for each of the motors is above 10%, preferably above 20% and most preferably above 25%.
[0161] The invention has been described above in detail with reference to embodiments thereof. However, as is readily understood by those skilled in the art, other embodiments are equally possible within the scope of the present invention, as defined by the appended claims.

Claims

1. A drive system for a patient lift, the drive system comprising: a drum configured to control a vertical movement of a patient support mounting device of the patient lift via a load bearing member; at least one motor adapted to drive the drum, each motor of the at least one motor being connected to a motor shaft gear via an output motor shaft; and a transmission connecting the at least one motor and the drum, the transmission being adapted to transfer torque from the at least one motor to the drum, wherein the transmission comprises a transmission interface adapted to interplay with the motor shaft gear, wherein the transmission interface is configured to receive the motor shaft gear in at least two configurations, each configuration being associated with an orientation of the output motor shaft relative the transmission interface, wherein the transmission interface comprises an input transmission gear adapted to interplay with the motor shaft gear, wherein the transmission comprises an output gear fixed to the drum, the output gear being connected to the transmission interface for receiving torque from the motor shaft gear, wherein the motor shaft gear and the input transmission gear form a worm drive, and wherein the output gear comprises a ring wheel fixed to the drum.
2. The drive system according to claim 1, wherein the output motor shaft has an orientation which is orthogonal to the drum in the at least two configurations.
3. The drive system according to claim 1, wherein the at least one motor comprises a first motor and a second motor, wherein the transmission interface is adapted to interplay with a first motor shaft gear connected to the first motor via a first output motor shaft and a second motor shaft gear connected to the second motor via a second output motor shaft.
4. A patient lift comprising the drive system according to claim 1, the patient support mounting device and the load bearing member, the patient support mounting device being connected to the drive system via the load bearing member.
5. A method for the drive system for the patient lift according to claim 3 to control the vertical movement of the patient support mounting device, the method comprising: obtaining a torque exerted by each of the first motor and the second motor; determining at least one torque differential value as a difference between the torque exerted by each of the first motor and the second motor; and adjusting the torque exerted by at least one of the first motor and the second motor to compensate for the determined at least one torque differential value.
6. The method according to claim 5, wherein the first motor and the second motor are operating at a same speed.
7. The method according to claim 5, further comprising, after the step of determining, a step of updating an Adjusted Power Level, APL, for each of the first motor and the second motor, and, the step of adjusting is performed by scaling the torque exerted by at least one of the first motor and the second motor with the APL associated with said at least one of the first motor and the second motor.
8. The method according to claim 5, wherein obtaining the torque exerted by each of the first motor and the second motor is based on an average current and a Pulse Width Modulation, PWM, duty cycle setting provided to control the respective motor.
9. The method according to claim 8, wherein the drive system is arranged with a speed limit, and wherein the method further comprises, before the step of adjusting: determining a target current and/or a target PWM duty cycle associated with the speed limit, and controlling at least one of the

first motor and the second motor to reach the target current and/or the target PWM duty cycle.

10. The method according to claim 5, wherein the method is repeated continuously, and wherein the adjusting is based on a control parameter comprising a product part, an integral part and a derivative part of the determined at least one torque differential value.

11. The method according to claim 5, wherein the step of determining further comprises determining which motor of the first motor and the second motor contributes the least torque, and wherein the step of adjusting comprises reducing the torque exerted by one of the first motor or the second motor to substantially the same torque as the torque contributed by the other one of the first motor or the second motor contributing the least torque.

12. A computer program product comprising instructions which, when executed by a control module, cause the control module to carry out the method of claim 5.